

Check Report — RRB + SSC Maths, 53 Formulas

Ye report sirf saboot ke liye hai. Padhai ke liye **RRB_SSC_Maths_Formulas.pdf** use karo.

Formula cards	53 (215 “sawaal mein ye ho → formula” lines)
RRB Mathematics (English)	5,213 sawaal — sab kisi card line se jude ✓
RRB Group D Maths (हिन्दी)	3,250 sawaal — sab jude ✓
SSC Maths Chapterwise (scan, OCR se padha)	5,034 sawaal alag kiye (book khud 5,037 batati hai) — sab jude ✓
Kul	13,497 sawaal
Card ke har example	215/215 computer se check ✓
RRB ke purane solved examples	1,235/1,235 book answer key se match ✓
RRB random test (pehle kabhi hal nahi kiye)	159/159 ✓ (2 mein book ne options nahi chhape — badle mein naye liye)
SSC random test (har card ke 3, page image se padh ke)	141/141 ✓ · 1 mein book ki apni galti

Sach-sach

- Har sawaal ko uske topic aur shabdon se ek card line par joda gaya hai; har ek ka alag se calculation nahi hua. Calculation se check hue: 1,235 RRB solved + 159 RRB random + 141 SSC random.
- SSC book scan (photo) thi, isliye text OCR se nikala. Card chunne wala computer SSC random test mein 137/142 baar wahi card chuna jo hal karte waqt sahi laga; baaki 5 bhi isi PDF ke kisi aur card se hal hue.
- Bilkul naye type ka sawaal aaye to PDF ke page 2 ki 5 trick lagao.

1. Kaun sa card kitne sawaal (coverage)

Chapter	Card	RRB EN	RRB HI	SSC
1. Number System	NS-1 BHAAG – SHESH	73	43	199
	NS-2 GINTI + FACTORS	65	34	27
	NS-3 DO NUMBER + SERIES	206	84	42
2. Decimal Fractions	DF-1 DECIMAL ↔ FRACTION	113	86	12
	DF-2 FRACTION: BADA-CHHOTA + JOD	130	48	45
3. Indices and Surds	IS-1 POWER (GHAAT)	51	37	13
	IS-2 √ ROOT + SURD	93	68	51
4. Simplification	SIM-1 BODMAS + “?”	167	178	200
	SIM-2 IDENTITY SHORTCUT	88	69	1
5. LCM & HCF	LH-1 HCF – LCM	126	74	57
	LH-2 SAATH KAB / SABSE BADA	104	54	31
6. Percentage	PC-1 % = HISSA	157	71	176
	PC-2 BADHA – GHATA (MULTIPLIER)	100	49	101
7. Profit & Loss	PL-1 CP → SP (PROFIT / LOSS)	112	64	204
	PL-2 GINTI · RATE · NAKLI BAAT	58	51	101
8. Discount	DS-1 DISCOUNT	72	77	177
9. Ratio & Proportion	RP-1 “1 HISSA” (RATIO)	110	75	120
	RP-2 CROSS GUNA (PROPORTION)	85	89	76
10. Partnership	PT-1 PROFIT = PAISA × SAMAY	37	22	44

11. Work & Time	WT-1 KAAM = LCM	170	82	220
	WT-2 LOG × DIN × GHANTE	35	25	38
12. Alligation / Mixture	MX-1 ALLIGATION (CROSS)	38	11	28
	MX-2 PAANI MILAO / NIKAALO	19	4	13
13. Pipe & Cistern	PP-1 NAL – TANKI	155	95	63
14. Simple Interest	SI-1 SIMPLE INTEREST (P·R·T)	202	78	98
15. Compound Interest	CI-1 COMPOUND INTEREST	143	54	98
	CI-2 CI – SI FARAK	61	14	59
16. Problems on Age	AG-1 UMAR (AGE)	238	122	38
17. Average	AV-1 AVERAGE = TOTAL ÷ n	117	82	214
18. Speed, Time & Distance	ST-1 DOORI = SPEED × SAMAY	106	47	129
	ST-2 AVERAGE + RELATIVE SPEED	80	42	63
19. Train	TR-1 TRAIN	125	69	36
20. Boat & Stream	BS-1 NAAV – DHAARA	47	13	34
21. Mensuration	MN-1 2-D AREA TABLE	220	131	187
	MN-2 3-D VOLUME TABLE	121	138	183
	MN-3 SIZE BADLA	18	12	30
22. Algebra	AL-1 IDENTITY: VALUE DAALO	114	98	379
	AL-2 EQUATION (LINEAR + QUADRATIC)	119	153	30
	AL-3 AP / GP (SERIES)	35	12	11
	AL-4 VENN (SETS)	24	0	0
23. Trigonometry	TG-1 θ DAALO + VALUE TABLE	130	97	402
	TG-2 ANGLE BADLO (90° JODI)	57	18	22
	TG-3 UNCHAI – DOORI	35	9	67
24. Co-ordinate Geometry	CG-1 POINTS: DOORI · BEECH · AREA	31	15	1
	CG-2 LINE + CIRCLE	18	2	11
25. Geometry	GE-1 KON (ANGLES)	174	162	176
	GE-2 TRIANGLE KI LAMBAI	42	34	176
	GE-3 CIRCLE: CHORD · TANGENT	60	12	229
26. Statistics & Probability	SP-1 MEAN – MEDIAN – MODE	176	109	3
	SP-2 RANGE · SD	64	8	0
	SP-3 PROBABILITY	24	3	0
27. Data Interpretation	DI-1 GRAPH / TABLE PADHO	222	177	319
28. Miscellaneous	MS-1 BACHA HISSA + EK KA KITNA	46	49	0
Total		5,213	3,250	5,034

2. RRB random test — 159/159 ✓

Har purane formula ke 3 sawaal random chune (seed 20261002) jo pehle kabhi hal nahi kiye the; sirf card se hal karke book key se milaye.

Card (line)	Sawaal	Value rakhi → answer	Book
NS-1 (5)	RRB EN Ch 1 Q297	$1 \Rightarrow 1^{102} \times 1^{52} \Rightarrow$ unit digit 1	✓ (b)
NS-1 (1)	RRB HI Ch 1 Q7	$4 \Rightarrow$ last 2 digits ÷ 4	✓ (b)

NS-1 (5)	RRB EN Ch 1 Q288	$1 \times 1 \Rightarrow 1$	✓ (c)
NS-2 (5)	RRB HI Ch 1 Q63	$1771 = 7 \times 11 \times 23 \Rightarrow 23 + 59 = 82 \Rightarrow 23 \times 59 = 1357$	✓ (a)
NS-2 (5)	RRB HI Ch 1 Q116	beech wala = $HCF(357, 609) = 21 \Rightarrow 17, 21, 29 \Rightarrow 67$	✓ (b)
NS-2 (2)	RRB HI Ch 1 Q146	odd $\Rightarrow 2$ hatao: $(5+1)(4+1)(6+1) = 210$	✓ (c)
NS-3 (4)	RRB EN Ch 1 Q70	options daalo: $25 \times 23 = 575$ ✓	✓ (c)
NS-3 (1)	RRB EN Ch 1 Q167	$8x + 4 = 100 \Rightarrow x = 12$	✓ (a)
NS-3 (1)	RRB HI Ch 1 Q92	$x + \frac{x}{3} = 12 \Rightarrow x = 12 \div \frac{4}{3} = 9$	✓ (b)
DF-1 (2)	RRB EN Ch 2 Q86	$12/30 = 2/5 \Rightarrow$ neeche sirf 5 $\Rightarrow$ terminating	✓ (b)
DF-1 (1)	RRB EN Ch 2 Q71	$4 + \frac{567-5}{990} = 4\frac{281}{495}$	✓ (a)
DF-1 (1)	RRB HI Ch 2 Q20	$\frac{7900-7}{9990} \Rightarrow 0.7\overline{900}$	✓ (c)
DF-2 (2)	RRB HI Ch 2 Q98	$\frac{5}{4} - \frac{3}{5} = \frac{13}{20}$	✓ (b)
DF-2 (2)	RRB HI Ch 2 Q99	$\frac{1}{5} - \frac{1}{12} = \frac{7}{60}$	✓ (c)
DF-2 (1)	RRB EN Ch 2 Q20	decimal: 0.2, 0.25, 0.125, 0.025 $\Rightarrow 25/1000$	✓ (b)
IS-1 (2)	RRB EN Ch 3 Q106	$\sqrt{\frac{45 \times 20}{12 \times 3}} = \sqrt{25} = 5$	✓ (d)
IS-1 (3)	RRB HI Ch 3 Q8	$x = \frac{\sqrt{455625}}{27} = 25 \Rightarrow \sqrt{x} = 5$	✓ (c)
IS-1 (3)	RRB EN Ch 3 Q82	$x^2 + 4x^2 = 625 \Rightarrow x = \sqrt{125}$	✓ (c)
IS-2 (2)	RRB EN Ch 3 Q97	$\frac{0.0013}{13} = 0.0001$	✓ (c)
IS-2 (1)	RRB HI Ch 3 Q34	last digit 4 $\Rightarrow 2/8$; $1831 \Rightarrow 42$ ($42 \times 43 = 1806 \leq 1831$) $\Rightarrow 428$	✓ (c)
IS-2 (3)	RRB HI Ch 3 Q86	$8\sqrt{5} = 17.84 \Rightarrow 11\sqrt{5} = 24.53$	✓ (b)
SIM-1 (4)	RRB EN Ch 4 Q153	$\frac{1}{3} \times \frac{7}{3} \times \frac{24}{5} \div \frac{16}{15} = \frac{7}{2}$	✓ (d)
SIM-1 (1)	RRB HI Ch 4 Q41	$56 \div 7 - 21 = -13$	✓ (c)
SIM-1 (4)	RRB EN Ch 4 Q173	of pehle: $\frac{11}{5} - \left(\frac{2}{5} - \frac{1}{5}\right) + \frac{3}{2} = \frac{7}{2}$	✓ (a)
SIM-2 (3)	RRB EN Ch 4 Q105	$\frac{(2\sqrt{2}+\sqrt{7})^2}{8-7} = 15 + 4\sqrt{14} \Rightarrow y = 4$	✓ (d)
SIM-2 (3)	RRB EN Ch 4 Q240	$\sum n^3 = 36^2 \Rightarrow (6^4)^{-5/2} = 6^{-10}$	✓ (c)
SIM-2 (3)	RRB EN Ch 4 Q206	$27 + 41 + 24 - 43 = 49$	✓ (c)
LH-1 (1)	RRB EN Ch 5 Q140	$4752 = 2^4 \cdot 3^3 \cdot 11 \Rightarrow$ teesra $= 2 \cdot 3^3 = 54$	✓ (a)

LH-1 (1)	RRB EN Ch 5 Q55	do prime $\Rightarrow$ HCF = 1	✓ (a)
LH-1 (2)	RRB EN Ch 5 Q127	$N = \frac{9 \times 180}{36} = 45$	✓ (c)
LH-2 (4)	RRB EN Ch 5 Q186	$\text{LCM}(3, 7) = 21 \text{ s} \Rightarrow 60 \div 21 \Rightarrow 2$	✓ (a)
LH-2 (1)	RRB EN Ch 5 Q214	$\text{LCM} = 4725 \Rightarrow 4725 - 3$	✓ (d)
LH-2 (1)	RRB HI Ch 5 Q19	$\text{LCM } 1001 \Rightarrow 9009 + 4$	✓ (d)
PC-1 (1)	RRB EN Ch 6 Q71	$\frac{40000}{200000} \times 100 = 20\%$	✓ (a)
PC-1 (4)	RRB EN Ch 6 Q240	Total = $6800 \div 34\% = 20000 \Rightarrow$ margin $(48 - 34)\% = 2800$, invalid $18\% = 3600$	✓ (b)
PC-1 (1)	RRB HI Ch 12 Q7	20% of 9000 g = 1800 g	✓ (a)
PC-2 (2)	RRB HI Ch 6 Q113	$2(10) + \frac{10^2}{100} = 21\%$	✓ (d)
PC-2 (3)	RRB EN Ch 8 Q24	$\frac{162000}{(0.9)^2} = 200000$	✓ (a)
PC-2 (2)	RRB EN Ch 6 Q233	$2(50) + \frac{50^2}{100} = 125\%$	✓ (c)
PL-1 (3)	RRB EN Ch 7 Q92	$CP = \frac{56 \times 100}{22.5 + 12.5} = 160 \Rightarrow SP = 160 \times 1.25 = 200$	✓ (d)
PL-1 (2)	RRB HI Ch 7 Q79	$2223 \times \frac{100}{117} = 1900$	✓ (d)
PL-1 (3)	RRB HI Ch 7 Q58	$CP \times \frac{28 - 11.6}{100} = 18 + 50 \times 1.116 \Rightarrow CP = 450$	✓ (c)
PL-2 (3)	RRB HI Ch 7 Q31	CP ₹8 / kg, SP ₹10 $\Rightarrow 25\%$	✓ (a)
PL-2 (5)	RRB EN Ch 7 Q165	chahiye $SP = 5 \times 1.4 = 7 \text{ lakh} \Rightarrow (7 - 1) \div 3 = 2 \text{ lakh}$	✓ (c)
PL-2 (1)	RRB HI Ch 7 Q28	$\frac{42 - 36}{36} \times 100 = 16\frac{2}{3}\% \text{ laabh}$	✓ (a)
DS-1 (1)	RRB HI Ch 8 Q77	$160 \times 0.88 = 140.80$	✓ (a)
DS-1 (3)	RRB HI Ch 8 Q47	$M = 118, D = 10 \Rightarrow 118 \times 0.9 - 100 = 6.2\%$	✓ (d)
DS-1 (1)	RRB EN Ch 8 Q52	$160 \times 0.88 = 140.80$	✓ (a)
RP-1 (4)	RRB EN Ch 9 Q160	$k = \frac{1920}{10 + 60 + 250} = 6 \Rightarrow 3k = 18$	✓ (c)
RP-1 (3)	RRB EN Ch 28 Q51	$G : B = 3 : 2 \Rightarrow \frac{2}{5} \times 3 \div 5 = \frac{6}{25}$	✓ (d)
RP-1 (2)	RRB EN Ch 9 Q69	Zinc barabar (21): 51 : 21 : 28	✓ (b)
RP-2 (2)	RRB HI Ch 9 Q78	$x = \frac{3 \times 8}{12} = 2$	✓ (b)
RP-2 (2)	RRB HI Ch 9 Q128	$4\text{th} = \frac{24 \times 45}{12} = 90$	✓ (a)
RP-2 (1)	RRB EN Ch 9 Q103	$k = \frac{3(-5) - 4(5)}{5 \cdot 4 - 7 \cdot 3} = 35 \Rightarrow 12k = 420$	✓ (b)

PT-1 (1)	RRB EN Ch 10 Q44	$10000 \times 12 : 15000 \times 8 = 1 : 1$	✓ (c)
PT-1 (1)	RRB EN Ch 10 Q35	$8000 \times 36 : 10000 \times 36 : 6000 \times 30 \Rightarrow Z = 9660 \times \frac{180}{828} = 2100$	✓ (b)
PT-1 (1)	RRB HI Ch 10 Q11	$3 : 2 : 5 \Rightarrow 1680, 1120, 2800$	✓ (c)
WT-1 (4)	RRB EN Ch 11 Q112	6 din mein $6 \left(\frac{1}{30} + \frac{1}{20} \right) = \frac{1}{2} \Rightarrow B: \frac{1}{2} \times 20 = 10$	✓ (c)
WT-1 (3)	RRB EN Ch 11 Q23	$k = 2, T = 17 \Rightarrow B = 17 \times 3 = 51$	✓ (b)
WT-1 (2)	RRB HI Ch 11 Q21	$\frac{1}{B} = \frac{1}{97.2} + \frac{1}{162} - \frac{1}{81} \Rightarrow 243$	✓ (a)
WT-2 (1)	RRB HI Ch 11 Q49	$7 \times 12 - 7 \times 5 = 49 \text{ man-days} \div 4 = 12\frac{1}{4}$	✓ (d)
WT-2 (1)	RRB EN Ch 11 Q81	$36m + 54b = 45m + 30b \Rightarrow m : b = 8 : 3 \Rightarrow \frac{18(16+9)}{32+6} = 11\frac{16}{19}$	✓ (a)
WT-2 (1)	RRB EN Ch 11 Q201	$\log \times \text{din const: } 105 \times 17 = 119 \times x \Rightarrow x = 15$	✓ (a)
MX-1 (2)	RRB EN Ch 12 Q49	1 hissa = $140 \div 4 = 35 \Rightarrow \text{copper } 3 \times 35$	✓ (c)
MX-1 (2)	RRB EN Ch 12 Q50	1 hissa = $90 \div 6 = 15 \Rightarrow 20 \times 15$	✓ (d)
MX-1 (2)	RRB EN Ch 12 Q56	$\frac{35.26}{1.25+2.015} = 10.80$	✓ (c)
MX-2 (1)	RRB EN Ch 12 Q7	acid fix: $2 \rightarrow 5 \text{ hisse} = 10 \text{ L} \Rightarrow 1 \text{ hissa } 10/3 \Rightarrow \text{acid } 10$	✓ (a)
MX-2 (1)	RRB HI Ch 12 Q13	doodh 6 fix $\Rightarrow \text{paani } 6 \times \frac{3}{2} = 9 \Rightarrow \text{jodo } 9 - 4 = 5$	✓ (c)
MX-2 (1)	RRB EN Ch 12 Q20	doodh $14 \div 25 = 56\%$	✓ (b)
PP-1 (4)	RRB EN Ch 13 Q79	aadha 2 h $\Rightarrow \text{baaki aadha } 4 \text{ tanker se } \frac{1}{2} \text{ h} \Rightarrow 2 \text{ h } 30 \text{ min}$	✓ (d)
PP-1 (4)	RRB EN Ch 13 Q93	P ne $24/32 = \frac{3}{4} \Rightarrow Q \text{ ne } \frac{1}{4} \Rightarrow \frac{1}{4} \times 48 = 12 \text{ min}$	✓ (a)
PP-1 (1)	RRB HI Ch 13 Q94	$\frac{1}{t} + \frac{2}{t} - \frac{1}{10} = \frac{1}{2.5} \Rightarrow t = 6$	✓ (a)
SI-1 (5)	RRB HI Ch 14 Q31	$P_1 : P_2 = \frac{1}{30} : \frac{1}{36} = 6 : 5 \Rightarrow 3600, 3000$	✓ (a)
SI-1 (1)	RRB EN Ch 14 Q152	$R = \frac{27 \times 100}{1080 \times \frac{4}{12}} = 10\%$	✓ (d)
SI-1 (6)	RRB HI Ch 14 Q75	$SI = 4500 \Rightarrow 4500 \div 18000 = 25\%$	✓ (c)
CI-1 (1)	RRB EN Ch 15 Q96	$8000(1.04^2 - 1) = 652.8$	✓ (a)
CI-1 (1)	RRB HI Ch 15 Q19	$y = \frac{612}{0.0816} = 7500 \Rightarrow A = 8112$	✓ (b)
CI-1 (2)	RRB EN Ch 15 Q177	$\left(\frac{7986}{6000} \right)^{1/3} = 1.1 \Rightarrow 10\%$	✓ (d)
CI-2 (1)	RRB EN Ch 15 Q120	$R = 100 \div 8 = 12.5\% \Rightarrow 8000(1.125^2 - 1) = 2125$	✓ (c)
CI-2 (1)	RRB EN Ch 15 Q110	half-yearly 5%, 2 baar: $P(0.05)^2 = 50 \Rightarrow P = 20000$	✓ (d)

CI-2 (1)	RRB EN Ch 15 Q111	$SI = \frac{CI}{1 + \frac{r}{100}} = 4400$	✓ (d)
AG-1 (2)	RRB EN Ch 16 Q169	$x = \frac{5(10-7)}{5-7-3-10} = 3 \Rightarrow \text{farak } 2x = 6$	✓ (b)
AG-1 (1)	RRB EN Ch 16 Q12	options daalo: $G = 15 \Rightarrow M = 75 \Rightarrow 78 = 4(18) + 6$ ✓	✓ (b)
AG-1 (3)	RRB HI Ch 16 Q45	abhi jod $28 + 2 \times 5 = 38 \Rightarrow \text{chhota } \frac{38-12}{2} = 13$	✓ (b)
AV-1 (5)	RRB EN Ch 17 Q113	$1050 - 360 - 126 = 564 \Rightarrow \div 3 = 188$	✓ (c)
AV-1 (4)	RRB EN Ch 17 Q32	$200 - 114 = 86 \Rightarrow \text{doosra max } 45 \Rightarrow 41$	✓ (a)
AV-1 (5)	RRB EN Ch 17 Q78	$625 - 300 - 220 = 105$	✓ (c)
ST-1 (3)	RRB EN Ch 18 Q148	$\frac{D/T}{(D/2)/(2T)} = 4 : 1$	✓ (d)
ST-1 (1)	RRB HI Ch 18 Q39	$450 \div 150 = 3 \text{ m/s} = 10.8 \text{ km/h}$	✓ (b)
ST-1 (3)	RRB EN Ch 18 Q2	car 66 $\Rightarrow$ bus $66 \times \frac{7}{11} = 42$	✓ (a)
ST-2 (1)	RRB HI Ch 18 Q63	$\frac{1}{\frac{1}{3} + \frac{1}{4} + \frac{5}{50}} = 33\frac{1}{3}$	✓ (b)
ST-2 (1)	RRB EN Ch 18 Q130	$\frac{2 \cdot x \cdot 2x}{3x} = 24 \Rightarrow x = 18$	✓ (a)
ST-2 (2)	RRB HI Ch 18 Q40	relative $1 \text{ km/h} \times 6 \text{ min} = 100 \text{ m} \Rightarrow 150 - 100 = 50$	✓ (b)
TR-1 (3)	RRB EN Ch 19 Q141	$S = \frac{195-155}{18-16} = 20 \text{ m/s} = 72 \text{ km/h}$	✓ (b)
TR-1 (2)	RRB HI Ch 19 Q10	$10 \text{ m/s} \times 53 = 530 \Rightarrow 530 - 110 = 420$	✓ (b)
TR-1 (3)	RRB HI Ch 19 Q29	$P = S(t - t_0): 200 = S \times 20 \Rightarrow S = 10 \Rightarrow L = 10 \times 20 = 200$	✓ (c)
BS-1 (3)	RRB HI Ch 20 Q5	$\frac{T}{T_0} = \frac{B^2}{B^2 - S^2} = \frac{16}{15} \Rightarrow B = 4S \Rightarrow 3S = 15 \Rightarrow S = 5$	✓ (d)
BS-1 (3)	RRB EN Ch 20 Q45	$\frac{d}{x-y} - \frac{d}{x+y} = t \Rightarrow d = \frac{(x^2 - y^2)t}{2y}$ (x = 5, y = 1, t = 1 daal ke option milao)	✓ (b)
BS-1 (2)	RRB EN Ch 20 Q16	$n = 2 \Rightarrow \frac{B}{S} = \frac{3}{1} \Rightarrow S = 9 \div 3 = 3$	✓ (b)
MN-1 (4)	RRB HI Ch 21 Q110	$2800 \times 15 \text{ cm}^2 = 4.2 \text{ m}^2 \times ₹25 = ₹105$	✓ (d)
MN-1 (3)	RRB EN Ch 21 Q198	max sides 35.5, 25.5 $\Rightarrow 2(61) = 122$	✓ (b)
MN-2 (5)	RRB HI Ch 21 Q202	volume same: $\frac{4}{3}\pi(6^3 - 4^3) = \pi 8^2 h \Rightarrow h = 3.17$	✓ (c)
MN-2 (2)	RRB HI Ch 21 Q182	$2\pi r(r + h) = 22(3.5 + h) = 269.5 \Rightarrow h = 8.75 \Rightarrow 8.75 : 7 = 5 : 4$	✓ (d)
MN-2 (2)	RRB EN Ch 21 Q287	$r = 10.5 \Rightarrow 2\pi r(r + h) = 66 \times 31.5 = 2079$	✓ (d)
MN-3 (1)	RRB EN Ch 21 Q334	$r_X l_X = 5r_Y(5l_X) \Rightarrow r_X = 25r_Y \Rightarrow \text{area } 625 : 1$	✓ (c)
MN-3 (1)	RRB HI Ch 21 Q183	$V \propto r^2 \Rightarrow 4 : 9$	✓ (c)

AL-1 (5)	RRB HI Ch 22 Q37	$x = 1, y = 2, z = 3$ daalo: $(0.5 + 6 - 1)^2 = 30.25 = \text{option (b)}$	✓ (b)
AL-1 (7)	RRB HI Ch 22 Q13	$x = 1, y = 2, z = 3$ daalo: $9 - 24 = -15 = \text{option (d)}$	✓ (d)
AL-1 (7)	RRB HI Ch 22 Q47	$x = 2$ daalo: $3 \times 1 \times 5 = 15 = 2^4 - 1$	✓ (d)
AL-2 (1)	RRB EN Ch 22 Q54	unique $\Leftrightarrow \frac{2}{2} \neq \frac{3}{p} \Rightarrow p \neq 3$	✓ (c)
AL-2 (2)	RRB HI Ch 22 Q241	$(a - b)^2 = S^2 - 4P = 25 + 24 = 49$	✓ (a)
AL-2 (2)	RRB HI Ch 22 Q248	simplify ke baad power: (a) 3, (b) $\sqrt{\quad}$ hata ke 2, (c) 1, (d) 1 $\Rightarrow$ (b)	✓ (b)
AL-3 (1)	RRB EN Ch 22 Q9	$5 + 8 \times 4 = 37$	✓ (b)
AL-3 (1)	RRB EN Ch 22 Q12	$2 + 9 \times 5 = 47$	✓ (c)
AL-3 (1)	RRB HI Ch 22 Q6	$3 \times 2^4 = 48$	✓ (a)
AL-4 (1)	RRB EN Ch 22 Q272	English: $30 + 20 + x = 60 \Rightarrow x = 10$ (Hindi: $25 + 20 + 10 = 55$ ✓)	✓ (c)
AL-4 (1)	RRB EN Ch 22 Q286	sirf $A = 33 - 10 - 8 + x = 21 \Rightarrow x = 6$	✓ (c)
AL-4 (1)	RRB EN Ch 22 Q285	$100 - 65 - 30 = 5\% \text{ of } 60 = 3$	✓ (b)
TG-1 (1)	RRB HI Ch 23 Q21	$\theta = 30^\circ$ daalo: $\frac{1}{64} + \frac{27}{64} = \frac{7}{16} = 1 - 3 \cdot \frac{1}{4} \cdot \frac{3}{4} \Rightarrow$ (a)	✓ (a)
TG-1 (1)	RRB EN Ch 23 Q108	$\theta = 30^\circ$ daalo $\Rightarrow 0$	✓ (c)
TG-1 (2)	RRB EN Ch 23 Q60	$\sec = \frac{k+1}{2} = \frac{(2-\sqrt{3})+(2+\sqrt{3})}{2} = 2 \Rightarrow \theta = 60^\circ \Rightarrow \frac{3}{2} + \frac{1}{2} = 2$ (book: $\sec = 2$ se $\theta = 60^\circ$ liya; strictly $\tan \theta$ negative hai)	✓ (b)
TG-2 (3)	RRB EN Ch 23 Q135	options daalo: $\cos 60^\circ = \frac{1}{2} = \sin 30^\circ \Rightarrow 30^\circ$	✓ (b)
TG-2 (3)	RRB EN Ch 23 Q16	$A = 60^\circ, B = 30^\circ$ daalo: $-3.73 + 3.73 = 0$	✓ (b)
TG-2 (1)	RRB EN Ch 23 Q24	$(48^\circ + \theta) + (42^\circ - \theta) = 90^\circ \Rightarrow \sin = \cos \Rightarrow 0$	✓ (d)
TG-3 (1)	RRB EN Ch 23 Q189	$d = 960 \times 1/60 = 16 \text{ km} \Rightarrow h = \frac{\sqrt{3}}{2} \times 16 = 13.86$	✓ (b)
TG-3 (1)	RRB EN Ch 23 Q183	same doori d: $x = d \tan 60^\circ, y = d \tan 30^\circ \Rightarrow 3 : 1$	✓ (a)
TG-3 (1)	RRB EN Ch 23 Q185	$45^\circ \Rightarrow h = d = 28.5 + \text{observer } 1.5 = 30$	✓ (a)
CG-1 (1)	RRB EN Ch 24 Q6	side 2, midpoint (2, 5), height $\sqrt{3} \Rightarrow (2 \pm \sqrt{3}, 5) \Rightarrow$ (c)	✓ (c)
CG-1 (1)	RRB HI Ch 24 Q15	$AB = 3\sqrt{5}, BC = 5\sqrt{5}, CA = 2\sqrt{5} \Rightarrow \frac{10+6+15}{30\sqrt{5}} = \frac{31\sqrt{5}}{150}$	✓ (a)
CG-1 (3)	RRB HI Ch 24 Q8	$\frac{1}{2} 0(3-1) + 2(1-2) + 3(2-3) = 2.5$	✓ (c)
CG-2 (1)	RRB EN Ch 24 Q2	intercept = $6 \div \text{coefficient}$: 2, $-3/2, -3$	✓ (d)
CG-2 (1)	RRB HI Ch 24 Q13	$D = 0 - 4 \cdot 9 \cdot 16 < 0 \Rightarrow$ x-axis ko na kaate na chhue	✓ (d)

CG-2 (1)	RRB EN Ch 24 Q37	$y = -4x + 2 \Rightarrow 4x + y = 2 \Rightarrow 2x + \frac{y}{2} = 1$	✓ (b)
GE-1 (5)	RRB EN Ch 25 Q101	diagonals bisect: $y + 2 = 3y \Rightarrow y = 1$; $x + 3 = 2x + 1 \Rightarrow x = 2$	✓ (a)
GE-1 (4)	RRB HI Ch 25 Q181	$180^\circ - \frac{360^\circ}{8} = 135^\circ$	✓ (d)
GE-1 (1)	RRB EN Ch 25 Q249	alternate kon barabar $\Rightarrow 35^\circ$	✓ (a)
GE-2 (5)	RRB HI Ch 25 Q98	$p = \sqrt{17^2 - 8^2} = 15$	✓ (a)
GE-2 (1)	RRB HI Ch 25 Q57	chinh milao: $Q \leftrightarrow Q, R \leftrightarrow P, S \leftrightarrow T \Rightarrow \Delta QPT$	✓ (b)
GE-2 (1)	RRB HI Ch 25 Q74	similar $\Rightarrow$ kon barabar $\Rightarrow 1 : 1$	✓ (b)
GE-3 (1)	RRB EN Ch 25 Q122	$EA \cdot EB = EC \cdot ED: 12 \times 3 = 4 \times EC \Rightarrow EC = 9 \Rightarrow CD = 5$	✓ (c)
GE-3 (2)	RRB HI Ch 25 Q167	tangent-chord kon = alternate segment ka kon $\Rightarrow 82^\circ$	✓ (d)
GE-3 (2)	RRB HI Ch 25 Q170	chord = r $\Rightarrow$ centre par $60^\circ \Rightarrow$ tangent se kon $30^\circ = \pi/6$	✓ (a)
SP-1 (1)	RRB HI Ch 26 Q7	$\sum fx = \bar{x} \sum f: 390 + 15X = 18(20 + X) \Rightarrow X = 10$	✓ (d)
SP-1 (4)	RRB HI Ch 26 Q94	Mean = $\frac{3(35)-27}{2} = 39$	✓ (b)
SP-1 (4)	RRB EN Ch 26 Q201	sort: 2,2,3,3,3,5,5,7,7,9 $\Rightarrow$ median 4, mode 3 $\Rightarrow 1$	✓ (b)
SP-2 (1)	RRB EN Ch 26 Q209	$CV = 6 \div 24 \times 100 = 25\%$	✓ (b)
SP-2 (1)	RRB EN Ch 26 Q228	$CV = 5 \div 11 \times 100 = 45.45\%$	✓ (a)
SP-2 (1)	RRB EN Ch 26 Q235	$\sigma = \sqrt{324} = 18$	✓ (b)
SP-3 (1)	RRB EN Ch 26 Q255	90 do-ank numbers, 21 prime $\Rightarrow 69/90 = 23/30$	✓ (b)
SP-3 (1)	RRB EN Ch 26 Q249	circle mein do saath: $\frac{2}{n-1} = \frac{2}{8} = \frac{1}{4}$	✓ (c)
SP-3 (1)	RRB EN Ch 26 Q264	$(3 + 5) \div 12 = 2/3$	✓ (a)
DI-1 (2)	RRB EN Ch 27 Q57	2017: 731, 2021: 1071 $\Rightarrow 731 \div 1071 \times 100 = 68\%$	✓ (c)
DI-1 (3)	RRB EN Ch 27 Q199	D1 men 430 $\rightarrow$ 240 $\Rightarrow$ 190 baante $\Rightarrow (250 + 350 + 190) : (440 + 180) = 79 : 62$ (graph par 450 chhapa, book ke hal mein 430)	✓ (d)
DI-1 (1)	RRB EN Ch 28 Q22	$(10 + 20)\%$ of 360 = 108	✓ (a)
MS-1 (4)	RRB HI Ch 28 Q27	3 9 1 9 0 9 1 6 8 $\Rightarrow$ aakhri 4: $9 + 1 + 6 + 8 = 24$	✓ (c)
MS-1 (4)	RRB HI Ch 28 Q40	2 4 6 2 4 2 0 $\Rightarrow 6 - 0 = 6$	✓ (b)
MS-1 (4)	RRB HI Ch 28 Q35	5 5 7 3 1 9 $\Rightarrow 5 + 1 = 6$	✓ (a)
MN-1 (3)	RRB EN Ch 21 Q190	doosri side $\sqrt{7.5^2 - 4.5^2} = 6 \Rightarrow$ area $4.5 \times 6 = 27$	✓ (a)

MN-3 (1)	RRB EN Ch 21 Q243	$\text{lambai} \times 3 \Rightarrow \text{volume} \times 3^3 = 27 \text{ guna}$	✓ (d)
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3. SSC random test — 141/141 ✓

Har card ke 3 sawaal random (seed 20261003); har sawaal book ke page ki image se padha, sirf card se hal kiya, book ke answer se milaya.

Card (line)	Sawaal	Value rakhi → answer	Book
NS-1 (1)	SSC Number System Q9 (p373)	$8 \rightarrow \text{last 3 digit } 46B \Rightarrow B = 4 \cdot 11 \rightarrow (8+5+4+4) - (A+1+6) = 0/11 \Rightarrow A = 3 \Rightarrow AB = 12$	✓ (b)
NS-1 (2)	SSC Number System Q187 (p398)	$a^n + b^n, n \text{ odd} \Rightarrow (31 + 43) = 74 = 2 \times 37 \text{ se kate} \Rightarrow \text{shesh } 0$	✓ (c)
NS-1 (1)	SSC Number System Q2 (p372)	$5^{71}(1 + 5 + 25 + 125 + 625) = 5^{71} \times 781 = 5^{71} \times 71 \times 11 \Rightarrow 71$	✓ (a)
NS-2 (5)	SSC Number System Q152 (p394)	11, 13, 17, 31, 37, 71, 73, 79, 97 $\Rightarrow 9$	✓ (b)
NS-2 (2)	SSC Number System Q90 (p386)	$9^9 = 3^{18} \Rightarrow \text{cube factors } \lfloor 18/3 \rfloor + 1 = 7$	✓ (b)
NS-2 (2)	SSC Number System Q157 (p395)	$72 = 2^3 \cdot 3^2 \Rightarrow 4 \times 3 = 12 \checkmark \cdot 20 \text{ odd ka jod } 20^2 = 400 \checkmark \cdot 97 \checkmark \Rightarrow \text{sab sahi}$	✓ (d)
NS-3 (1)	SSC Number System Q257 (p407)	$S = 11, x^2 + y^2 = 65 \Rightarrow P = (121 - 65)/2 = 28 \Rightarrow x^3 + y^3 = S^3 - 3PS = 1331 - 924 = 407$	✓ (b)
SP-1 (2)	SSC Number System Q275 (p409)	sort: 0,0,0,1,1,2,2,2,2,3,3,4,5,5,5 $\Rightarrow 8\text{th} = 2$ (median)	✓ (d)
AV-1 (2)	SSC Number System Q244 (p406)	Total = avg $\times n = 0 \Rightarrow \text{ek positive baaki sab ko balance kar sakta} \Rightarrow 1088 - 1 = 1087$	✓ (b)
DF-1 (3)	SSC Decimal and Fraction Q29 (p416)	$y = 9/16 - 0.0625 = 1/2 \Rightarrow x + z = 37/24 - 1/2 = 25/24$	✓ (c)
DF-1 (3)	SSC Decimal and Fraction Q22 (p415)	$0.1101 + 0.300909 = 0.411009$	✓ (a)
DF-1 (3)	SSC Decimal and Fraction Q28 (p416)	I: $0.15 \times 4 = 0.6 \checkmark$ · II: $0.01 + 0.0001 + 0.000001 = 0.010101 \checkmark$	✓ (d)
DF-2 (3)	SSC Decimal and Fraction Q13 (p413)	do cancel series: $10/11 + (1/3)(1 - 1/31) = 10/11 + 10/31 = 420/341$	✓ (d)
DF-2 (1)	SSC Decimal and Fraction Q2 (p411)	I: $11.5 + 17.75 - 5.2 - 2.1 = 21.95 = 439/20 \checkmark$ · II, III cross guna se galat	✓ (a)
DF-2 (1)	SSC Decimal and Fraction Q1 (p411)	decimal bana ke: $0.042 < 0.055 < 0.071 \checkmark \cdot 0.081 > 0.076 > 0.072 \checkmark$	✓ (c)
IS-1 (1)	SSC Indices and Surds Q22 (p421)	options daalo: $\sqrt{x} = 2 \Rightarrow 21^2 + 20^2 = 29^2 \checkmark \Rightarrow x = 4$	✓ (c)
IS-1 (1)	SSC Indices and Surds Q19 (p420)	same base: 3 ki power $6 - 2a = -26 \Rightarrow a = 16 \Rightarrow \sqrt{(16 + 9)} = 5$	✓ (b)
IS-1 (1)	SSC Indices and Surds Q18 (p420)	$3^2 + 4^2 = 5^2 \Rightarrow \sqrt[4]{x} = 2 \Rightarrow x = 16$	✓ (a)
IS-2 (5)	SSC Indices and Surds Q64 (p428)	$2^{25} \cdot 3^{21} \cdot 5^8$ common: $P \rightarrow 16, Q \rightarrow 4, R \rightarrow 6, S \rightarrow 15 \Rightarrow P > S > R > Q$	✓ (a)
IS-2 (2)	SSC Indices and Surds Q8 (p419)	$\sqrt{(256/729)} + \sqrt{(1/81)} = 16/27 + 3/27 = 19/27$	✓ (c)
IS-2 (3)	SSC Indices and Surds Q54 (p427)	rationalise: $23/(\sqrt{2} + \sqrt{3} + \sqrt{6}) = 4.11 \Rightarrow 8.66 + 9.90 - 2.45 - 4.11 = 12$	✓ (c)

SIM-1 (4)	SSC Simplification Q195 (p476)	BODMAS: $68 \div (25 - 8) = 4$	✓ (d)
SIM-1 (2)	SSC Simplification Q34 (p447)	of pehle: $135 \div 5/3 = 81$, $81 - 2 + 6 = 85$ · neeche $228 - 8 = 220 \Rightarrow 85/220 = 17/44$	✓ (c)
SIM-1 (2)	SSC Simplification Q216 (p479)	$1.00250625 \div 0.9525 = 1.0525$	✓ (d)
DF-2 (3)	SSC Simplification Q208 (p478)	cancel series: $\frac{1}{4}(1/3 - 1/903) = 25/301$	✓ (c)
DF-2 (3)	SSC Simplification Q198 (p476)	$25.3 \times 24.7 = 624.91 \Rightarrow (a^3 + b^3)/[100(a^2 - ab + b^2)] = (a + b)/100 = 0.05 = 5 \times 10^{-2}$	✓ (a)
SIM-2 (3)	SSC Simplification Q205 (p477)	$1/4 + 1/6 + \dots + 1/40 = \frac{1}{2}(1/2 + \dots + 1/20) = (k - 1)/2$	✓ (c)
LH-1 (3)	SSC LCM and HCF Q62 (p437)	$LCM = h \cdot a \cdot b = 8 \times 7 \times 13 = 728$	✓ (d)
LH-1 (2)	SSC LCM and HCF Q82 (p439)	$81 \times 7 = 567$, $81 \times 8 = 648 \Rightarrow 1/567 + 1/648 = 5/1512$	✓ (a)
LH-1 (1)	SSC LCM and HCF Q58 (p436)	$xy = HCF \times LCM = 12 \times 2448 \Rightarrow (x + y)^2 = 60^2 + 4xy \Rightarrow 348$	✓ (c)
LH-2 (1)	SSC LCM and HCF Q11 (p430)	$LCM = 3600 \Rightarrow 3600k + 4$, 7 se kate $\Rightarrow k = 5 \Rightarrow 18004 \Rightarrow$ hazaar ka digit 8	✓ (d)
LH-2 (1)	SSC LCM and HCF Q15 (p430)	$LCM 180 \Rightarrow 180k + 9$, 11 se kate $\Rightarrow 1089 \Rightarrow 1 + 0 + 8 + 9 = 18$	✓ (c)
LH-2 (1)	SSC LCM and HCF Q20 (p431)	$LCM 420 \Rightarrow 420k + 1$, 2800-3000 $\Rightarrow 2941 \Rightarrow 16$	✓ (b)
PC-1 (3)	SSC Percentage Q127 (p561)	bacha = $I \times 0.42 \times 0.45 = 4725 \Rightarrow I = 25000 \Rightarrow$ education = $0.55 \times 0.42 \times 25000 = 5775$	✓ (d)
PC-1 (5)	SSC Percentage Q264 (p583)	$1.06x + 1.04(70000 - x) = 73520 \Rightarrow x = 36000 \Rightarrow 36000 - 34000 = 2000$	✓ (d)
PC-1 (1)	SSC Percentage Q123 (p561)	$12.5\% \times 30\% \times 1440 = 54$	✓ (d)
PC-1 (5)	SSC Percentage Q13 (p546)	literate $2125 -$ male $1265 = 860 \Rightarrow 860/1125 \times 100 = 76 \frac{4}{9} \%$	✓ (d)
PC-2 (1)	SSC Percentage Q111 (p559)	$D = 100 \Rightarrow C 110$, $B 88$, $A 110 \Rightarrow B$, A se 20% kam (22% nahi) $\Rightarrow$ (a) galat	✓ (a)
PC-2 (4)	SSC Percentage Q71 (p554)	$110(100 - x) = 50(100 + x) \Rightarrow x = 37.5 \Rightarrow 243.75$ vs $57.5\% \times 180 = 103.5 \Rightarrow 135.5\% \approx 136\%$	✓ (c)
PL-1 (2)	SSC Profit and Loss Q175 (p612)	$CP = 800 + 350 = 1150 \Rightarrow SP = 1150 \times 120/100 = 1380$	✓ (b)
PL-1 (2)	SSC Profit and Loss Q277 (p630)	$5400 \times 0.7 \times 1.6 = 6048 \Rightarrow (6048 - 5400)/5400 = 12\%$ gain	✓ (b)
PL-1 (2)	SSC Profit and Loss Q293 (p632)	$CP : SP = 100 : 125 = 4 : 5$	✓ (b)
PL-2 (5)	SSC Profit and Loss Q8 (p587)	$CP: 60 \times 5/4 + 60 \times 3/2 = 165$ · $SP: 120 \times 4/3 = 160 \Rightarrow$ loss $5/165 = 3 \frac{1}{33} \%$	✓ (b)
PL-2 (1)	SSC Profit and Loss Q147 (p608)	loss = 3 ka CP $\Rightarrow CP = 2520 \div (21 - 3) = 140$	✓ (a)
PL-1 (5)	SSC Profit and Loss Q228 (p621)	$MP/CP = 125/87.5 = 1.43 \Rightarrow 43\%$ upar	✓ (b)
DS-1 (3)	SSC Discount Q93 (p650)	$MP = CP \times (100 + 20)/(100 - 16) = 11900 \times 120/84 = 17000$	✓ (d)
DS-1 (1)	SSC Discount Q9 (p636)	$7600 \times 0.9 \times (1 - d) = 5814 \Rightarrow 1 - d = 0.85 \Rightarrow 15\%$	✓ (c)
DS-1 (2)	SSC Discount Q49 (p642)	$20, 20 \Rightarrow 20 + 20 - 4 = 36\% = 331.20 \Rightarrow MP = 920$	✓ (b)
RP-1 (2)	SSC Ratio and Proportion Q18 (p517)	B barabar (35): $55 : 35 : 133$	✓ (a)
RP-1 (4)	SSC Ratio and Proportion Q163 (p537)	$k = S(d - c)/(ad - bc) = 4000(3 - 2)/(9 - 8) = 4000 \Rightarrow B = 4k = 16000$	✓ (c)
RP-1 (1)	SSC Ratio and Proportion Q187 (p541)	temporary = $3/8 \cdot 2/3 + 2/5 \cdot 1/3 = 23/60 \Rightarrow$ permanent $37/60 = 740 \Rightarrow T 1200 \Rightarrow 560 - 160 = 400$	✓ (a)
RP-2 (1)	SSC Ratio and Proportion Q100 (p527)	cross guna: $(52 - x)(19 - x) = (47 - x)(20 - x) \Rightarrow x = 12 \Rightarrow$ mean of 25, 4 = 10	✓ (b)

RP-1 (1)	SSC Ratio and Proportion Q183 (p540)	$k^2 = 1029 \div (1 + 4 + 16) = 49 \Rightarrow k = 7 \Rightarrow 4k - k = 21$	✓ (a)
RP-1 (1)	SSC Ratio and Proportion Q96 (p527)	$x = 90 \times 3/5 = 54, y = 36 \Rightarrow 18$	✓ (c)
PT-1 (1)	SSC Partnership Q20 (p718)	$24000 \times 8 : 28000 \times 10 : 20000 \times 12 = 24 : 35 : 30 \Rightarrow 25810 \times 24/89 = 6960$	✓ (a)
PT-1 (1)	SSC Partnership Q30 (p720)	70% baanta, $A = 5/13 \Rightarrow x \times 0.7 \times 5/13 = 87500 \Rightarrow x = 325000$	✓ (d)
PT-1 (1)	SSC Partnership Q14 (p717)	60% of 60000 = 36000 $\Rightarrow A = 36000 \times 3/10 = 10800$	✓ (c)
WT-1 (3)	SSC Time and Work Q234 (p772)	$A : B : C = 15 : 10 : 6 \Rightarrow \text{kaam} = 31 \times 10 = 310 \Rightarrow 150\% = 465 \Rightarrow 465 \div 15 = 31$	✓ (c)
WT-1 (2)	SSC Time and Work Q52 (p738)	LCM 180: 5, 3, 4 $\Rightarrow$ teeno 6 $\Rightarrow B = 6 - 4 = 2 \Rightarrow 180 \div 2 = 90$	✓ (c)
WT-1 (4)	SSC Time and Work Q160 (p759)	A, 2 din pehle chhode $\Rightarrow (8 + 2) \times 12 \div (8 + 12) = 6$	✓ (a)
WT-2 (1)	SSC Time and Work Q69 (p742)	$1 \times 12 \times 10 = 1 \times 8 \times x \Rightarrow x = 15$	✓ (a)
WT-2 (1)	SSC Time and Work Q177 (p763)	1 man = 2 women $\cdot 96 - 48 = 48 \text{ man-days} \div (6 + 2) = 6$	✓ (b)
WT-2 (1)	SSC Time and Work Q21 (p733)	$M_1 D_1 / W_1 = M_2 D_2 / W_2: 15 \times 6/10 = x \times 10/50 \Rightarrow x = 45$	✓ (a)
MX-1 (2)	SSC Alligation Q14 (p725)	apple 65% aur 66%, barabar quantity $\Rightarrow$ average = 65 5/6 %	✓ (b)
MX-1 (2)	SSC Alligation Q24 (p726)	acid 20x + 30y, paani 80x + 170y $\Rightarrow$ cross guna $91(20x + 30y) = 19(80x + 170y) \Rightarrow 300x = 500y \Rightarrow 5 : 3$	✓ (a)
MX-1 (1)	SSC Alligation Q9 (p724)	$25(8x - 10)/8 = 100 \Rightarrow 8x - 10 = 32 \Rightarrow \text{kul } 42 \text{ L}$	✓ (a)
MX-2 (1)	SSC Alligation Q16 (p725)	options daalo: $x = 3 \Rightarrow$ juice 2.5, paani $1.5 + 1 = 2.5 \Rightarrow 1 : 1$ ✓	✓ (a)
MX-2 (1)	SSC Alligation Q13 (p725)	paani fix 18 L = 75% $\Rightarrow$ naya kul 24 L $\Rightarrow$ acid 4 L jodo	✓ (d)
MX-2 (1)	SSC Alligation Q11 (p724)	acid fix 2.5 L = 20% $\Rightarrow$ kul 12.5 L $\Rightarrow$ 5 L paani jodo	✓ (d)
PP-1 (1)	SSC Pipes and Cisterns Q8 (p778)	$A : B : C = 1 : 3 : 6 \Rightarrow \text{kaam} = 10 \times 4 = 40 \Rightarrow A \text{ akela } 40 \text{ h}$	✓ (b)
PP-1 (1)	SSC Pipes and Cisterns Q34 (p783)	3 h mein $3(1/10 + 1/15 + 1/30) = 3/5$ bhara $\Rightarrow D \text{ } 3/5 \text{ ko } 30 \text{ h} \Rightarrow$ poora 50 h	✓ (d)
PP-1 (4)	SSC Pipes and Cisterns Q54 (p787)	10 h: $10(1/16 + 1/24 - 1/40) = 19/24 \Rightarrow$ baaki $5/24 \div (1/24 - 1/40) = 12\frac{1}{2} \text{ h}$	✓ (d)
SI-1 (2)	SSC Simple Interest Q71 (p676)	1 saal SI = $(600 - 500)/2 = 50 \Rightarrow P = 500 - 5 \times 50 = 250$	✓ (d)
SI-1 (1)	SSC Simple Interest Q5 (p664)	$T = SI \times 100 \div (P \times R) = 2725 \times 100 \div (5450 \times 12.5) = 4$	✓ (d)
SI-1 (2)	SSC Simple Interest Q72 (p676)	same R, T $\Rightarrow SI \propto P: 2646 \times 9600/8400 = 3024$	✓ (a)
CI-1 (1)	SSC Compound Interest Q11 (p684)	$(1 + R)^3 = 12042/8028 = 1.5 \Rightarrow P = 8028 \div 1.5 = 5352$	✓ (a)
CI-1 (1)	SSC Compound Interest Q66 (p695)	$10000 \times 1.12^2 \times (1 + 0.12 \times \frac{3}{4}) - 10000 = 3673$	✓ (c)
CI-1 (1)	SSC Compound Interest Q155 (p714)	lagataar saal: $(2409 - 2190) \div 2190 \times 100 = 10\%$	✓ (b)
CI-2 (1)	SSC Compound Interest Q83 (p698)	2 saal: $D = P(R/100)^2 \Rightarrow P = 60.50 \div 0.11^2 = 5000$	✓ (d)
CI-2 (1)	SSC Compound Interest Q100 (p701)	$SI = 25000 \times 11 \times 2/100 = 5500 \cdot CI = 25000 \times 21\% = 5250 \Rightarrow$ Varun 250 zyada	✓ (c)

CI-2 (1)	SSC Compound Interest Q72 (p696)	$SI = 1920 \cdot CI = 8000 \times 1.1^2 \times 1.04 - 8000 = 2067.20 \Rightarrow 147.20$	✓ (a)
AG-1 (1)	SSC Age Problems Q20 (p842)	$t = 14; x = 14(5 - 6)/(4 \cdot 6 - 5 \cdot 5) = 14 \Rightarrow 56, 70 \Rightarrow 5 \text{ saal baad } 68 : 82 = 34 : 41$	✓ (a)
AG-1 (1)	SSC Age Problems Q2 (p839)	$x = 8(11 - 12)/(9 \cdot 12 - 10 \cdot 11) = 4 \Rightarrow 36, 40 \Rightarrow 3 \text{ saal baad } 39 + 43 = 82$	✓ (c)
AG-1 (1)	SSC Age Problems Q15 (p841)	ek saal pehle se $t = 4: x = 4(6 - 5)/(4 \cdot 5 - 3 \cdot 6) = 2 \Rightarrow 8, 6 \Rightarrow +10 \Rightarrow 18 : 16 = 9 : 8$	✓ (c)
AV-1 (2)	SSC Average Q62 (p490)	sahi avg $= 68 + (130 - 112)/40 = 68.45$	✓ (d)
AV-1 (3)	SSC Average Q141 (p503)	$(32 \times 58 + 48 \times 52) \div 80 = 54.4$	✓ (b)
AV-1 (6)	SSC Average Q186 (p510)	$A = 5.3 \times 50 = 265 \Rightarrow B \text{ ko } 5 \text{ over mein } 7.2 \times 5 = 36 \text{ chahiye} \Rightarrow \text{abhi } 265 - 36 = 229$	✓ (d)
ST-1 (1)	SSC Speed, Time and Distance Q129 (p811)	Meenal 700 m @5 $\Rightarrow$ samay 140 $\Rightarrow$ Nitu 980 m $\Rightarrow$ jeet 220 m (book ka hal bhi 220; option mein nahi)	book galti
ST-1 (1)	SSC Train Q19 (p820)	car $486 \div 9 = 54 \Rightarrow$ train $54 \times 13/6 = 117 \Rightarrow 6 \text{ h mein } 702 \text{ km}$	✓ (b)
ST-1 (1)	SSC Speed, Time and Distance Q14 (p792)	$0.9 \text{ km} \div 0.1 \text{ h} = 9 \text{ km/h}$	✓ (a)
ST-1 (4)	SSC Train Q50 (p828)	$(80 - 72) \div 80 \times 60 = 6 \text{ min}$	✓ (a)
ST-1	SSC Speed, Time and Distance Q25 (p793)	$1800 \div 5 = 360 \text{ km/h} \times 5/18 = 100 \text{ m/s}$	✓ (d)
ST-2 (1)	SSC Speed, Time and Distance Q149 (p815)	$D = 80, T = 1: 25\% = 20 \text{ km in } \frac{1}{3} \text{ h} \Rightarrow 60 \text{ km/h}$	✓ (c)
TR-1 (3)	SSC Train Q65 (p830)	$L = 60S \Rightarrow (L + 2L) \div S = 180 \text{ s}$	✓ (d)
TR-1 (4)	SSC Train Q66 (p831)	$S_1 + S_2 = 320/9 \cdot S_1 = (160 + 200)/18 = 20 \Rightarrow S_2 = 140/9 \Rightarrow 9 : 7$	✓ (b)
TR-1 (1)	SSC Train Q67 (p831)	$t = L \div S \Rightarrow (6/3) : (5/2) = 4 : 5$	✓ (b)
BS-1 (1)	SSC Boat and Stream Q16 (p835)	$B = (D + U)/2 = (30 + 20)/2 = 25$	✓ (c)
ST-2 (1)	SSC Boat and Stream Q34 (p838)	aana-jaana: $2xy/(x + y) = 2 \times 18 \times 30 \div 48 = 22.5$	✓ (a)
BS-1 (1)	SSC Boat and Stream Q22 (p836)	$S = (D - U)/2 = (23 - 11)/2 = 6$	✓ (a)
MN-1 (3)	SSC Mensuration 2D Q164 (p329)	$x \times 4x = 1764 \Rightarrow x = 21 \Rightarrow \text{chaudai } 84$	✓ (b)
MN-1 (4)	SSC Mensuration 2D Q115 (p320)	bahar path: $(a + 9)^2 - a^2 = 252 \Rightarrow 18a + 81 = 252 \Rightarrow a = 9.5$	✓ (a)
MN-1 (5)	SSC Mensuration 2D Q72 (p312)	$2\pi(9.5 - 6.7) = 2 \times 22/7 \times 2.8 = 17.6$	✓ (d)
MN-2 (6)	SSC Mensuration 3D Q194 (p369)	$\frac{1}{3} \times (\sqrt{3}/4 \times 36) \times h = 45\sqrt{3} \Rightarrow 3\sqrt{3} h = 45\sqrt{3} \Rightarrow h = 15$	✓ (b)
MN-3 (1)	SSC Mensuration 3D Q19 (p339)	volume $a^3 : b^3 = 64 : 27 \Rightarrow \text{side } 4 : 3$	✓ (a)
MN-2 (1)	SSC Mensuration 3D Q39 (p341)	$TSA = (l + b + h)^2 - \text{diagonal}^2 = 400 - 144 = 256$	✓ (c)
MN-1 (1)	SSC Mensuration 2D Q44 (p306)	$54x = 1080 \Rightarrow 240, 340, 500 \cdot \text{Heron: } s = 540 \Rightarrow 36000 \text{ m}^2 = 3.6 \text{ ha}$	✓ (a)
MN-3 (1)	SSC Mensuration 3D Q78 (p347)	$V \propto r^2 h \Rightarrow 1.02^3 - 1 = 6.12\%$	✓ (c)
MN-1 (3)	SSC Mensuration 2D Q64 (p310)	area $= \frac{1}{2}bh \Rightarrow h_1/h_2 = (1/2) \times (4/3) = 2/3$	✓ (d)
AL-1 (7)	SSC Algebra Q338 (p66)	$(a - b)^3 = a^3 - 3a^2b + 3ab^2 - b^3 \Rightarrow x \text{ wala term } 3 \times 3x \times 16 = 144x$	✓ (c)
AL-1 (6)	SSC Algebra Q132 (p32)	$\frac{1}{2}(a + b + c)[(a - b)^2 + (b - c)^2 + (c - a)^2] = 190.5 \times 24 = 4572$	✓ (d)

AL-1 (7)	SSC Algebra Q273 (p56)	$x + 1/x = 3 \cdot \text{upar-neeche} \div x: (x^3 + 1/x^3) \div (x + 1/x + 5) = (27 - 9) \div 8 = 2\frac{1}{4}$	✓ (c)
AL-2 (1)	SSC Algebra Q8 (p10)	teen equation $\Rightarrow a, b, c$ (Cramer) $\Rightarrow a + c = 5/12$	✓ (b)
AL-2 (1)	SSC Algebra Q15 (p11)	bada $(27 + 3)/2 = 15$, chhota $12 \Rightarrow 5 : 4$	✓ (d)
AL-2 (1)	SSC Algebra Q9 (p11)	$x = 4y \Rightarrow 6y = 24 \Rightarrow y = 4, x = 16 \Rightarrow 44/20 = 11/5$	✓ (b)
AL-3 (1)	SSC Number System Q230 (p404)	AP 1, 4, 7 ... 20 term $= 590 \cdot 1/(1 \cdot 2) + 1/(2 \cdot 3) + \dots = 20/21 \Rightarrow 12410/21$	✓ (a)
NS-2 (1)	SSC Number System Q239 (p405)	5 ke multiple $200/5 = 40$, unme 20 ke 10 $\Rightarrow 30$	✓ (b)
AL-3 (1)	SSC Number System Q235 (p404)	pehle n odd numbers ka jod $= n^2$	✓ (b)
TG-2	SSC Trigonometry Q269 (p130)	$51^\circ = 90^\circ - 39^\circ: \sin^2 51 + \sin^2 39 = 1, \tan^2 51 = \cot^2 39 = x^2 - 1, \text{aakhri} = 1 \Rightarrow x^2 - 1$	✓ (c)
TG-1 (1)	SSC Trigonometry Q57 (p91)	$\tan + \cot = k \Rightarrow \tan^2 + \cot^2 = k^2 - 2 = 36 - 2 = 34$	✓ (d)
TG-1 (1)	SSC Trigonometry Q137 (p106)	$\theta = 45^\circ$ daalo: upar $2(\frac{1}{4}) - 3(\frac{1}{2}) = -1$, neeche $\frac{1}{4} - \frac{1}{4} - 1 = -1 \Rightarrow 1$	✓ (a)
TG-1	SSC Trigonometry Q33 (p86)	$\sin^2 = 1 - \cos^2 \Rightarrow 3\cos^2 + \cos - 2 = 0 \Rightarrow \cos \theta = 2/3 \Rightarrow (\cos + 1)/\sin = (5/3)/(\sqrt{5}/3) = \sqrt{5}$	✓ (c)
TG-2 (1)	SSC Trigonometry Q255 (p127)	$x = 30^\circ$ daalo: $0.3248 - 0.1083 = 0.2165 = \sin 120^\circ/4 \Rightarrow (d)$	✓ (d)
TG-2 (3)	SSC Trigonometry Q281 (p132)	table: $A + B = 60^\circ, A - B = 30^\circ \Rightarrow A = 45^\circ, B = 15^\circ$	✓ (b)
TG-3 (1)	SSC Height and Distance Q10 (p159)	dori l: $h = l \sin \theta \Rightarrow l = 90 \div \sin 30^\circ = 180$	✓ (a)
TG-3 (1)	SSC Height and Distance Q17 (p160)	$h = 16 \times \sin 60^\circ = 8\sqrt{3}$	✓ (d)
TG-3 (1)	SSC Height and Distance Q24 (p162)	$h = d \div (\cot 45^\circ - \cot 60^\circ) = 100 \div (1 - 1/\sqrt{3}) = 50(3 + \sqrt{3})$	✓ (a)
CG-1 (2)	SSC Co-ordinate Geometry Q2 (p296)	line $y = k$ mein reflection: $(x, 2k - y) = (5, 6 + 3) = (5, 9)$	✓ (d)
CG-2 (1)	SSC Co-ordinate Geometry Q10 (p297)	$y - y_1 = m(x - x_1): y + 5 = 2(x - 3) \Rightarrow 2x - y - 11 = 0$	✓ (c)
CG-2 (1)	SSC Co-ordinate Geometry Q8 (p297)	y-axis par $x = 0 \Rightarrow (0, 6) \cdot$ x-axis par $y = 0 \Rightarrow (2.5, 0) \Rightarrow 8.5 = 17/2$	✓ (d)
CG-2 (3)	SSC Co-ordinate Geometry Q9 (p297)	$(x - 0)^2 + (y - 0)^2 = 6^2 \Rightarrow x^2 + y^2 - 36 = 0$	✓ (b)
GE-1 (3)	SSC Geometry Q144 (p203)	bahar ke bisectors: $\angle BOC = 90^\circ - A/2 = 57^\circ \Rightarrow A = 66^\circ$	✓ (a)
GE-2	SSC Geometry Q49 (p182)	BPT: $AD/AB = AE/AC \Rightarrow 2/7.2 = x/9 \Rightarrow x = 2.5$	✓ (d)
GE-3 (2)	SSC Geometry Q559 (p291)	tangents ka kon + centre ka kon $= 180^\circ \Rightarrow 180 - 130 = 50^\circ$	✓ (c)
GE-2 (2)	SSC Geometry Q91 (p192)	centroid: teeno chhote triangle barabar $\Rightarrow ABG = \frac{1}{3} ABC$	✓ (c)
GE-2 (2)	SSC Geometry Q569 (p293)	equilateral mein O, G, I, H — chaaron ek hi point	✓ (b)
GE-2 (2)	SSC Geometry Q544 (p288)	tangent–chord kon $45^\circ \Rightarrow \angle AOB = 90^\circ \Rightarrow \triangle NOB$ right $\Rightarrow$ hypotenuse ka median $= \frac{1}{2} \times 10 = 5$	✓ (b)
GE-1	SSC Geometry Q416 (p259)	cyclic PQRS: $\angle PQR + \angle PSR = 180^\circ \Rightarrow 180 - 95 = 85^\circ$	✓ (a)
GE-3 (2)	SSC Geometry Q502 (p279)	bahar se do tangent: $\angle APO = \angle BPO = 35^\circ$	✓ (a)
GE-3 (1)	SSC Geometry Q422 (p261)	$\angle APB = \frac{1}{2} \times 64 = 32^\circ \cdot$ semicircle mein $\angle PAQ = 90^\circ \Rightarrow \angle AXP = 180 - 90 - 32 = 58^\circ$	✓ (c)
DI-1 (3)	SSC Data Interpretation Q122 (p874)	$(48 + 36) : (56 + 35) = 84 : 91 = 12 : 13$	✓ (d)
DI-1 (4)	SSC Data Interpretation Q104 (p869)	R ka avg $(285 + 288 + 294)/3 = 289 \cdot Q 2015 = 320 \Rightarrow x = 31 \Rightarrow 30$ aur 35 ke beech	✓ (c)
DI-1 (2)	SSC Data Interpretation Q95 (p867)	2017: $C + D = 52.5 + 60 = 112.5 \cdot$ 2016: $A + B + C = 25 + 40 + 45 = 110 \Rightarrow 2.27\% \approx 2.3\%$	✓ (b)

4. Books ki galtiyan (jo check karte waqt mili)

Book	Sawaal	Kya galat hai
RRB EN	Ch 10 Q1	English book mein A ki capital ₹50,000 chhapi hai (galat); Hindi book (HI2) mein ₹4,50,000 hai $\Rightarrow 450000 \times 12 : 9B = 3 : 2 \Rightarrow B = ₹4,00,000 = \text{option (c)}$.
RRB EN	Ch 12 Q42	1.5% aur 2.5% ko milakar 3.5% ban hi nahi sakta (mean hamesha dono ke beech). Question galat; book ka 1 : 2 cross rule ko ulta lagane se aaya.
RRB EN	Ch 12 Q45	Book ne ₹160 aur ₹400 (SP) ko hi rates maan kar mean ₹240 se alligation lagaya $\Rightarrow 2 : 1$. Loss/profit se asli CP (₹200, ₹333.3) lein to 7 : 3 aata hai — option mein nahi.
RRB EN	Ch 13 Q119	Book ne 2 ghante ka net $\frac{1}{4}$ maan kar $4 \times 2 = 8$ h likha. Par 5th ghante mein A akela $\frac{1}{2}$ se poora bhar deta hai $\Rightarrow$ sahi $4 + \frac{3}{4} = 4.75$ h = option (a).
RRB HI	Ch 13 Q79	EN119 jaisa: sahi uttar 4.75 घंटे (a); book ka (d) 8 घंटे aakhri ghante ka trap bhool gaya.
RRB EN	Ch 13 Q59	Wording ulti chhapi hai. Sahi arth: doosra pehle se 5 h KAM, teesre se 4 h ZYADA leta $\Rightarrow 15, 10, 6$ ($1/15 + 1/10 = 1/6$) $\Rightarrow 15 \checkmark$.
RRB EN	Ch 18 Q59	Question mein “reduced his speed to 2 km/h” chhapa hai, par book ka answer (12:30 pm) “by 2 km/h” (yaani 4 km/h) se aata hai: $18/6 + 12/4 = 6$ h. “to 2 km/h” lene par 3:30 pm aata jo options mein nahi.
RRB EN	Ch 19 Q39	Answer key mein (c) 700 m chhapa hai, par book ki apni solution bhi $L = 800$ m nikalti hai (55 m/s ki train). Sahi option (a) 800 m — Hindi book (HI38) mein bhi (a).
RRB EN	Ch 23 Q60	$\sec \theta + \tan \theta = 2 - \sqrt{3}$ se $\tan \theta$ negative aata hai (strictly answer -1 , option mein nahi); book ne $\sec \theta = 2 \Rightarrow \theta = 60^\circ$ maan ke (b) 2 diya
RRB EN	Ch 27 Q199	graph par District 1 men 450 chhapa hai, book ke hal mein 430; 430 se (d) 79 : 62 aata hai (450 se 81 : 62, option mein nahi)
SSC	Speed, Time and Distance Q129	Book ke apne hal mein bhi 220 m aata hai, par 220 option mein nahi; key ne (a) 200 m diya.